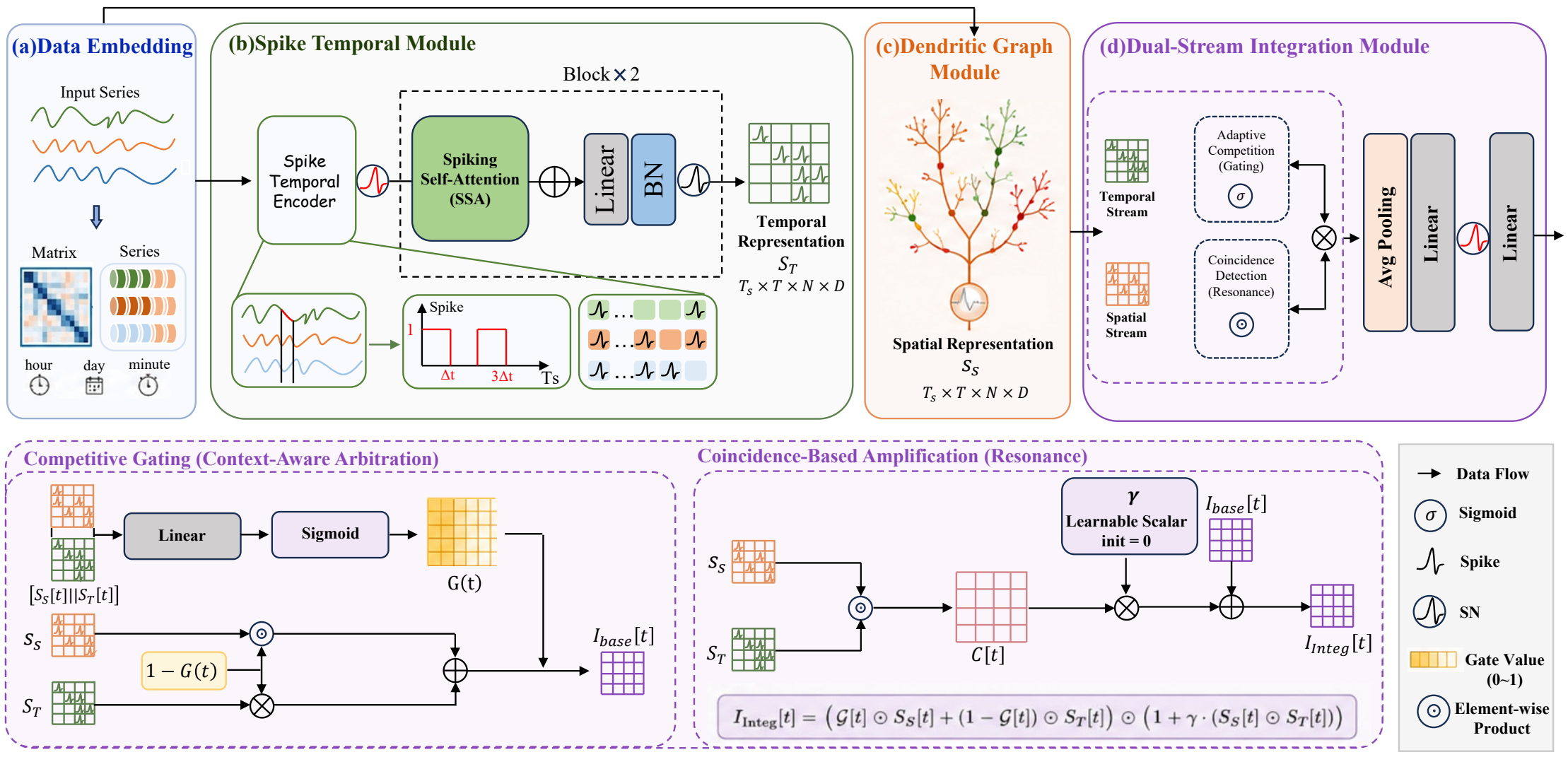

(a)Data Embedding
Input Series
Matrix
Series
hour
day
minute
(b)Spike Temporal Module
Block × 2
Spike Temporal Encoder
Spiking Self-Attention (SSA)
Linear
BN
Temporal Representation
S_T
$T_s \times T \times N \times D$
Spike
1
Δt
3Δt
Ts
(c)Dendritic Graph Module
Spatial Representation
S_S
$T_s \times T \times N \times D$
(d)Dual-Stream Integration Module
Temporal Stream
Spatial Stream
Adaptive Competition (Gating)
Coincidence Detection (Resonance)
Avg Pooling
Linear
Linear
Competitive Gating (Context-Aware Arbitration)
Linear
Sigmoid
$[S_S[t]||S_T[t]]$
G(t)
S_S
$1 - G(t)$
S_T
$I_{base}[t]$
Coincidence-Based Amplification (Resonance)
γ
Learnable Scalar
init = 0
$I_{base}[t]$
S_S
S_T
$C[t]$
$I_{Integ}[t]$
$I_{\text{Integ}}[t] = \left(\mathcal{G}[t] \odot S_S[t] + (1 - \mathcal{G}[t]) \odot S_T[t]\right) \odot \left(1 + \gamma \cdot (S_S[t] \odot S_T[t])\right)$
Data Flow
Sigmoid
Spike
SN
Gate Value (0~1)
Element-wise Product